HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND
COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS
AND STATISTICS

F72ZA1
FINANCIAL MATHEMATICS 1

10 December 2003

Time allowed 2 hours

Attempt all questions

Total Marks 60

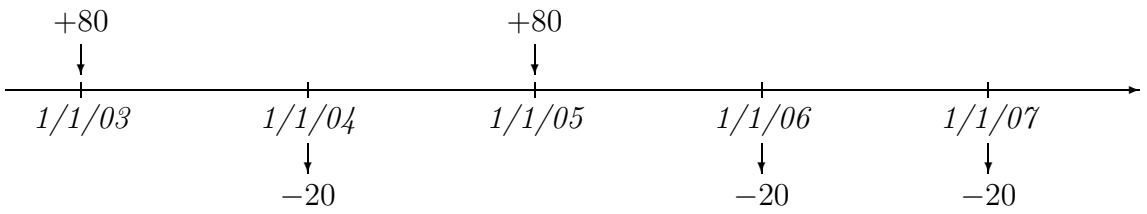
Only approved calculators may be used.

1. Fill in the missing values in the following table:

i	v	d	$i^{(4)}$	$d^{(12)}$
0.10	?	?	?	?

[4 marks]

2. The effective rate of interest will be 5.5% per annum before 1 July 2004, and 3% per half-year thereafter. A series of cashflows is given as follows:



   (a) What is the accumulation on 1 July 2007 of these cashflows?

   (b) What is the value on 1 April 2005 of these cashflows?

[8 marks]

3. Evaluate the following:

   (a) $\ddot{s}_{\overline{10}|}^{0.1}$.

   (b) $a_{\overline{\infty}|}^{(2)\ 0.065}$.

   (c) $(Is)_{\overline{10}|}^{0.1}$.

[6 marks]

continued ...

4. Define accumulation and discounting factors as follows:

$$\begin{aligned} A(s,t) &= \text{the accumulation at time } t \text{ of a cashflow of 1 at time } s \quad (s \le t) \\ V(s,t) &= \text{the present value at time } s \text{ of a cashflow of 1 at time } t \quad (s \le t). \end{aligned}$$

You may assume the following properties:

$$\begin{aligned} A(s,t) &= A(s,r)A(r,t) \qquad (s \le r \le t) \\ V(s,t) &= A(s,t)^{-1} \qquad (s \le t). \end{aligned}$$

Suppose that cashflows occur at times $t_1, t_2, \ldots, t_n$. The cashflow at time t_r is of amount c_r $(r = 1, 2, \ldots, n)$.

(a) Let the present value at time $s < t_1$ of these cashflows be X. Write down an expression for X.

(b) Let the accumulation at time $t > t_n$ of these cashflows be Y. Write down an expression for Y.

(c) Prove that $Y = XA(s,t)$.

[6 marks]

5. (a) You borrow £90,000 to buy a house. The loan is to be repaid over 25 years by level annual repayments, and the interest rate is 6% per annum effective.

(i) What is the level annual repayment?

(ii) What is the capital outstanding just after the 10th repayment has been made?

[4 marks]

(b) Just after the 10th repayment has been made, you ask the lender if you can miss the next three annual repayments. The lender agrees, but will charge a rate of interest of 8% per annum effective during the period when payments are suspended. What is the new annual repayment when you resume making payments?

[4 marks]

[Total 8 marks]

6. What is the price at issue per £100 nominal, of a bond with term 25 years, with coupons of 8% per annum payable each quarter-year in arrear, redeemable at 103%, if the yield obtained by an investor subject to income tax at 20% and capital gains tax at 25% is 12% per annum effective?

[7 marks]

continued ...

7. An investment fund, which has no assets on 31 December 2001, receives inflows of £1,000 on 1 January in 2002, 2003 and 2004, and on 31 December 2004 it has assets of £3,641.

   (a) Show that the money-weighted rate of return during the years 2002–2004 is 10% per annum.

   [1 mark]

   (b) Suppose that the fund has assets of £2,152.5 on 31 December 2003. Show that the linked internal rate of return during the years 2002–2004 is 8.39% per annum, based on the subdivision of the period into 2002–2003 and 2004.

   [4 marks]

   (c) Suppose that the fund has assets of £1,070 on 31 December 2002. Show that the time weighted rate of return during the years 2002–2004 is 8.72% per annum.

   [2 marks]

   [Total 7 marks]

8. (a) A mining company plans to set up and operate a new mine. It will take two years to set up, at a cost of £200 million at the start of the first year. At the end of each of the second to tenth years inclusive, it will make a profit of £80 million, after which it will close.

   (i) What is the Net Present Value (NPV) of the project, at a discount rate of 15% per annum?

   (ii) What is the breakeven period of the project, at a discount rate of 15% per annum?

   (iii) Find the yield on this project.

   [10 marks]

   (b) Suppose that there are cost over-runs in setting the mine up, and the mine becomes more difficult to work as time passes, so that it in fact costs £220.5 million to set up, and the profit at the end of the second year is the expected £80 million, but then profits decrease by 5% each year until the mine is closed at the end of the tenth year. What is the yield on the project now? [Hint: $220.5 = 200(1.05)^2$].

   [4 marks]

   [Total 14 marks]

Total 60 marks